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Offer

Offers- No: 160246 / 4
date: 31/03/2017

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Inquiry dated 25/01/2017
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**EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND
TURNING CENTRE VSC 315 DUO KBU
for straight and angular ball grooves (AC- and UF-joints)
for the simultaneous soft-milling of 2 Outer Races of the same
type**



Summary

1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 DUO KBU

1.1 BASIC MACHINE

1.1.15 1 PRICE OF MACHINE VSC 315 DUO KBU (SIEMENS)

1.2 ADDITIONAL EQUIPMENT

1.2.2 1 WORKPIECE LOADING by motorised, chain-type conveyor

1.2.3 1 LINEAR GANTRY LP3, WITH ROTATION MODULE

1.2.4 1 POSITION DETECTION

1.2.5 1 GRIPPER JAWS OUTER RACE CLAMPING DIA. 20-36 (SPARE PART)

1.2.6 2 AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK

1.2.7 1 SHUTTLE ZR120DUO L-R/ L-L FOR WORKPIECE LOADING

1.2.8 2 WORKPIECE CARRIER RIGID OUTER RACE AC21021-C8001

1.2.9 2 WORKPIECE CARRIER RIGID OUTER RACE AC21022-C8001

1.2.10 2 WORKPIECE CARRIER RIGID OUTER RACE AC21023-C8001

1.2.11 2 WORKPIECE CARRIER RIGID OUTER RACE AC21024-C9002

1.2.12 1 SHUTTLE ZR120DUO L-R/ L-L FOR WORKPIECE UNLOADING

1.2.13 1 PRISM FINISH PART FOR OUTER RACES

1.2.14 1 SAFETY ENCLOSURE

1.2.15 2 SAFETY SWITCH WITH CABLE FOR SAFETY ENCLOSURE

1.2.16 1 MINI MANUAL CONTROL UNIT (SIEMENS)

1.2.17 1 CONNECTING KIT FOR MANUAL CONTROL UNIT (SIEMENS)

1.2.19 1 SWARF CONVEYOR - SIDE DISCHARGE

1.2.20 1 SWARF CONVEYOR, CHAIN TYPE

1.2.21 1 CHIP BOX (LARGE VERSION)

1.2.22 1 AIR GUN

1.2.23 2 DRY-DUST AND MIST EXTRACTOR

1.2.25 1 EMAG REMOTE EXPERTS

1.2.26 1 NETWORK DRIVE CONNECTION

1.2.27 1 EMERGENCY RETRACTION AT POWER FAILURE

1.2.28 1 INTERFACE TO EXTERNAL HANDLING SYSTEM (LH), SIEMENS

Price does not include options (*)

Offer : 160246 / 4

Hyundai WIA Corp.

31/03/2017

2/28

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- 1.2.29 1 INTERFACE TO EXTERNAL HANDLING SYSTEM (RH), SIEMENS
- 1.2.30 1 WORKSHOP-ORIENTATED PROGRAMMING AC AND UF-JOINTS
- 1.2.32 1 SPARE PARTS PACKAGE
- 1.2.33 1 ASIAN PACKAGE
- 1.2.34 1 3-COLOUR STATUS LIGHT (Q-LIGHT)
- 1.2.35 1 SPECIAL MACHINE COLOUR

1.3 COMPONENT RELATED EQUIPMENT

- 1.3.2 1 SET OF MILLING TOOLHOLDERS HSK 63-C FOR ROUGH MILLING
- 1.3.3 1 SET OF MILLING TOOLHOLDERS HSK 63-C FOR FINISH MILLING
- 1.3.4 1 SET OF MILLING TOOLS FOR SOFT MILLING OF OUTER RACE AC21021-C8001
- 1.3.5 1 SET OF MILLING TOOLS FOR SOFT MILLING OF OUTER RACE AC21022-C8001
- 1.3.6 1 SET OF MILLING TOOLS FOR SOFT MILLING OF OUTER RACE AC21023-C8001
- 1.3.7 1 SET OF MILLING TOOLS FOR SOFT MILLING OF OUTER RACE AC21024-C9002
- 1.3.8 1 EQUIPMENT FOR BALL TRACK MILLING MACHINE
- 1.3.9 1 SETTING DIE FOR MILLING HEAD EXCHANGE
- 1.3.11 2 ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER SIN85 - with 1 medium duct
- 1.3.12 2 WORKPIECE POSITION MONITORING, (AIR SENSING)
- 1.3.13 2 COLLET CHUCK 120 (HAINBUCH)
- 1.3.14 2 SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (HAINBUCH) AC 21021-C8001
- 1.3.15 2 SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (HAINBUCH) AC 21022-C8001
- 1.3.16 2 SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (HAINBUCH) AC 21023-C8001
- 1.3.17 2 SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (HAINBUCH) AC 21024-C9002
- 1.3.18 1 CHANGING FIXTURE FOR COLLET CHUCK 120

1.4 SERVICES

- 1.4.2 1 PROJECT ENGINEERING
- 1.4.3 1 TOOLING LAYOUT FOR OUTER RACE AC21021-C8001
- 1.4.4 1 TOOLING LAYOUT FOR OUTER RACE AC21022-C8001
- 1.4.5 1 TOOLING LAYOUT FOR OUTER RACE AC21023-C8001
- 1.4.6 1 TOOLING LAYOUT FOR OUTER RACE AC21024-C9002

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- 1.4.7 1 MACHINE SETUP FOR OUTER RACE AC21021-C8001
- 1.4.8 1 MACHINE SETUP FOR OUTER RACE AC21022-C8001
- 1.4.9 1 MACHINE SETUP FOR OUTER RACE AC21023-C8001
- 1.4.10 1 MACHINE SETUP FOR OUTER RACE AC21024-C9002
- 1.4.11 1 PRE-ACCEPTANCE OF THE MACHINE
- 1.4.12 1 INSTALLATION AND DEMONSTRATION (OVERSEAS)
- 1.4.13 1 FINAL ACCEPTANCE AT CUSTOMER'S PLANT AC21021-C8001
- 1.4.14 1 FINAL ACCEPTANCE AT CUSTOMER'S PLANT AC21022-C8001
- 1.4.15 1 FINAL ACCEPTANCE AT CUSTOMER'S PLANT AC21023-C8001
- 1.4.16 1 FINAL ACCEPTANCE AT CUSTOMER'S PLANT AC21024-C9002
- 1.4.19 1 PRODUCTION SUPPORT 1 WEEK AFTER START OF PRODUCTION
- 1.4.22 5 ADDITIONAL SET OF DOCUMENTATION
- 1.4.23 2 DOCUMENTATION ON CD
- 1.4.25 1 COMPLIANCE SPECIFICATION SHEET / DELIVERY CONDITIONS
- 1.4.27 1 OPERATING COURSE (Basic training)
- 1.4.28 1 MAINTENANCE (MECHANICAL AND ELECTRICAL) SERVICE COURSE (Basic training)
- 1.4.30 1 PACKING AND TRANSPORTATION OF TEST PIECE AND INSPECTION GAUGE FROM INDIA
- 1.4.31 1 PACKING AND TRANSPORTATION OF TEST PIECE AND INSPECTION GAUGE FROM KOREA
- 1.4.32 1 FOB DELIVERY FREE GERMAN SEAPORT
- 1.5 MACHINE SPECIFICATIONS**

Total: 1.223.850,00 €
Discount: 373.850,00 €
Final Price: 850.000,00 €

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4/28



1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 DUO KBU

1.1 BASIC MACHINE

- 1.1.1 1 MINERALIT (CAST GRANITE) MACHINE BED
consisting of two separate sections, to prevent interaction during the machining process. For transport purposes the two sections are mechanically coupled.
The machine bed is manufactured from high quality Mineralit (cast granite) which guarantees superior thermal stability and excellent dampening characteristics. Mineralit has excellent vibration dampening properties which are six to eight times better than cast iron. The large distance between the two high precision pre-stressed linear roller guideways of the X-axis guarantees high accuracy and high dynamics. The guideways are outside the working area and therefore require no protective covers.
- 1.1.2 1 ENERGY CONTAINER WITH CONTROL CABINET
Electrical cabinet, cooled with water/air heat exchanger, equipped to VDE 0113 and EN 13849 and integrated in the over-all machine frame, thus ensuring a totally transportable unit. 400 V / 50 Hz machine voltage.
Other voltage on request.
The central grease supply system for lubrication of the recirculating ballscrews and guideway systems and also the service unit for compressed air supply are integrated in the energy container.
- 1.1.3 1 CNC-CONTROL SIEMENS
According to control specification.
- 1.1.4 1 HYDRAULIC UNIT
The hydraulic unit is equipped with a self regulating pump.
Supply 15 l/min (at 70 bar)
Tank capacity 63 l
- 1.1.5 1 COOLING UNIT (15 KW, water - air)
A cooling unit controls the thermal stability of the machine. Spindle bearings, drive unit and motor spindle for milling head are maintained at ambient temperature (10°C - 40°C).
Cooling capacity: 15 kW
- 1.1.6 1 MACHINE ENCLOSURE
The machining area is effectively separated from the load/unload area by the two side walls of the Mineralit machine base and a sliding door to the pick-up station.
A large door at the front of the machine provides access to the machining area (one for each spindle). The door features a safety glass window to DIN EN ISO 23125:2010 specification.
A inspection lamp is installed in the machining area.

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5/28

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- 1.1.7 2 MULTI-FUNCTIONAL CROSS-SLIDE WITH Y-AXIS
For the complete machining with driven tools or with drilling, milling, reaming or grinding spindles.
Max. Y-axis stroke 315 mm.
This unit is equipped with three axes (X, Y and Z) for each spindle. Both spindles can be programmed independently via the CNC control.
High rapid feed enable short idle times by component load/unload and gauging (optionally).
The axes drives are quick reaction frequency controlled service free servomotors which drive the slides over high precision ballscrew spindles. The X-, Y- and Z-axis are equipped with completely enclosed linear measuring systems.
A hydraulically operated counterbalance compensates the weight of the vertical spindle. On activation of the emergency-stop or in case of power failure, a brake will stop the vertical axis.
- 1.1.8 2 MOTOR SPINDLE
Spindle headstock unit with motor spindle
The motor spindle is a high dynamic frequency controlled service free AC synchronous spindle motor.
High degree of spindle rigidity due to triple bearing support over a short distance.
Precision shoulder bearings arranged in tandem (double-O). Additional bearing at spindle end for length compensation. All bearings are lifetime lubricated and thus maintenance free.
Sealing air to prevent dirt from penetrating from the outside.
The guideway system in the Z axis is equipped with hydrostatic bearings for compensation the cutting forces which guarantee excellent guideway quality by minimum friction.
Thermo-symmetrical design and integrated coolant system guarantee constantly maintained accuracy.
Positioning accuracy of spindle: +/- 0,01 angular degrees (C-axis)
Electro-hydraulic spindle locking optionally.
- 1.1.9 2 MILLING HEAD FOR MILLING OF BALL GROOVES
with 2 milling spindles
consisting of:
one axis, NC controlled swivelling unit (B-axis)
Swivelling housing guided by roller bearings in the side walls of the machine base.
Swivelling angle -30° to +90°
Swivelling drive by torque drive. Direct measuring system.
- 2 MOTOR SPINDLE HPC190 (manually tool clamping)
- | | | |
|-----------------------------|-----|-----------|
| Max. power 40 % ED | kW | 15 |
| Torque at 40 % ED | Nm | 67 |
| Max. spindle speed | rpm | 6.000 |
| Full power at spindle speed | rpm | 3.000 |
| Spindle taper DIN 69893 | HSK | C63 / F80 |
- Fluid-cooled spindle housing. Hybrid bearing.
Lifetime lubricated. Temperature controlled.
Rotating components are precision-balanced in two planes.
Coolant supply to tool holder pocket with minimal quantity lubrication max. 10 bar.
Pneumatic spindle nose cleaning device. Air seal between spindle and housing.
(warranty period 18 months)

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6/28

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- 1.1.10 1 EMAG OPERATING AND SETTING MASKS
The EMAG masks offer the setter/ operator optimal support in setting and operating the machine.
Multiple base- and monitoring functions including data input are, depending on the machine type, available.
- 1.1.11 1 EMAG - REMOTE DIAGNOSIS - SIEMENS
Control system: Sinumerik 840Dsl
- Linkage with user network necessary
- Connection via Internet and safe VPN technology to EMAG server
Scope of supply:
- Software for remote maintenance
- No additional equipment is supplied
Content:
- Remote access to NC / PLC / operating system for detailed error diagnosis and the possibility to support the customer in case of queries or problems.
Preconditions:
-Tele-diagnostics via Internet with VPN.
Your advantages:
- Quick, cost-effective support, resulting in reduced downtimes.
- Up-to-date problem solving without lengthy delays.
- Program extensions can be realised at a reduced cost.
- "On-the-spot" co-operation between specialist and operating/service personnel allows for complex errors to be diagnosed, analysed and eliminated quickly.
Use of the equipment past the warranty expiration date will be charged.
Option:
- EMAG REMOTE EXPERTS
- 1.1.12 1 For simple and fast programming of the machine on an appropriate ball track, the machine can be equipped with workshop-oriented programming modules. These modules are not included in the basic machine price and are described separately in the chapter „options to the CNC control“.
On the other hand, the part programs can also be completed by the operator in normal ISO code.
- 1.1.13 1 STANDARD COLOUR
Machine - light grey to RAL 7035,
front door - grey to RAL 7046,
automation, swarf conveyor - grey to RAL 7046.
- 1.1.14 1 NOISE LEVEL
A maximum sound level of 78 dB(A) to DIN 45635 is obtained when turning with a continuous cut on a C45 steel disc in both longitudinal and transverse direction. If the workpiece has for example an interrupted cut due to its geometry or its clamping condition there may occur a higher sound level.
For an extra cost sound deadening material can be fitted to reduce this higher sound level.
- 1.1.15 1 PRICE OF MACHINE VSC 315 DUO KBU (SIEMENS)

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7/28

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1.2 ADDITIONAL EQUIPMENT

1.2.1 COMPONENT STORAGE AND TRANSPORT SYSTEM

- 1.2.2 1 WORKPIECE LOADING by motorised, chain-type conveyor
Supplied by costumer.
Please note:
The hight of the infeed conveyor have to be 1050 [mm].
The infeed conveyor should be provided with height-adjustable feet.
The workpiece stop at the end of the infeed conveyor must be adjustable too.

- 1.2.3 1 LINEAR GANTRY LP3, WITH ROTATION MODULE
Type: LP03-0630-0500-E-S-D-C LE
2-axis linear gantry
Drive by NC servomotor. Direct connection to the machine.
Energy guiding on all axes via cable track.
Rotation unit driven by a servo motor with rotary inlet.
Workpiece transport with a pneumatic 3-finger centric gripper
incl. 1 set of workpiece-specific gripper jaws.
without position detection and sensor gripper monitoring.

Swivel angle	± 360°
X-axis stroke	630 mm
Z-axis stroke	500 mm
Workpiece weight	max. 3,5kg

- 1.2.4 1 POSITION DETECTION
Position detection of the workpiece via Keyence laser sensor, mounted on the gantry frame LP3

- 1.2.5 1 GRIPPER JAWS OUTER RACE CLAMPING DIA. 20-36 (SPARE PART)
1 set of gripper jaws for 3-finger centric part gripper

- 1.2.6 2 AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK
For the cleaning of the pick-up unit and the clamping chuck by air blast. The air is processed by a service unit and directed at the component through nozzles.

- 1.2.7 1 SHUTTLE ZR120DUO L-R/ L-L FOR WORKPIECE LOADING
Type: ZR120-A-1100-700-LE-B-1
Type: ZR120DUO-A-1600-1305-800-LE-B-1
For the workpiece transport through the machine, with a basic pallet for blank parts.
Consists of a spring-loaded pick-up station, including detection and emergency stop.
Slip / brake plate coated with spring steel, 1 trough for dry processing.
Mounting kit on the machine base body, 1x light barrier factory. Balluff.
Driven via servomotor.
The pick-up spindle grasps the blank parts of the shuttle.
The workpieces are positioned NC-controlled for the pick-up spindle. Different workpiece diameters and heights are defined in the NC part program.

- 1.2.8 2 WORKPIECE CARRIER RIGID OUTER RACE AC21021-C8001
for the location of outer races. Designed for one workpiece.

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8/28

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- 1.2.9 2 WORKPIECE CARRIER RIGID OUTER RACE AC21022-C8001
for the location of outer races. Designed for one workpiece.
- 1.2.10 2 WORKPIECE CARRIER RIGID OUTER RACE AC21023-C8001
for the location of outer races. Designed for one workpiece.
- 1.2.11 2 WORKPIECE CARRIER RIGID OUTER RACE AC21024-C9002
for the location of outer races. Designed for one workpiece.
- 1.2.12 1 SHUTTLE ZR120DUO L-R/ L-L FOR WORKPIECE UNLOADING
Type: ZR120DUO-A-1600-1305-800-LE-B-1
For the workpiece transport of the parts by the machine, without pick-up station.
Slip / brake plate coated with spring steel, 1 trough for dry processing.
Mounting kit on the machine base body, 1x light barrier factory. Balluff.
Driven via servomotor.
The pick-up spindle is used to deposit the parts on the shuttle, which then (depending on
the program selection), to the finished part storage or to the SPC removal.
Without shuttle prisms for pushing out the workpieces.
Please note:
**The height of the shuttle is 1050 [mm]. The height of the storage unit for finished
parts must also be 1050 [mm]**
The storage unit must be provided with height-adjustable feet.
- 1.2.13 1 PRISM FINISH PART FOR OUTER RACES
Prism for pushing the finished parts onto the outstocker.
- 1.2.14 1 SAFETY ENCLOSURE
Safety enclosure supplied by customer.
Please note:
- EMAG will supply an incomplete / unsafe machine.
- The customer is responsible for the safety of the system (machine and periphery).
- 1.2.15 2 SAFETY SWITCH WITH CABLE FOR SAFETY ENCLOSURE
for customer supplied safety enclosure
consisting of:
- cable
- closing mechanism
- 1.2.16 1 MINI MANUAL CONTROL UNIT (SIEMENS)
For operating a combined positioning and transfer unit and/or a linear gantry loader.
In the setting up mode of the automation components: the shuttle, the vertical and
horizontal axis of the gantry, also the swivelling and gripping function of the part gripper
can be manually operated with the manual control unit.
If the doors of the safety enclosure are open, the acknowledgment key must be pressed
on the manual control unit.
- 1.2.17 1 CONNECTING KIT FOR MANUAL CONTROL UNIT (SIEMENS)
(at operator panel)

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1.2.18 COOLANT AND SWarf DISPOSAL SYSTEM

- 1.2.19 1 SWarf CONVEYOR - SIDE DISCHARGE
Special designed:
Swarf exit height 260 mm. Located to the left of the machine.
- 1.2.20 1 SWarf CONVEYOR, CHAIN TYPE
Swarf exit height approx. 1250 mm. Located to the rear of the machine.
- 1.2.21 1 CHIP BOX (LARGE VERSION)
with caster
- 1.2.22 1 AIR GUN
with hose
- 1.2.23 2 DRY-DUST AND MIST EXTRACTOR
For dry hard machining.
Including stainless steel separator baffle. High-precision filtration unit with filter inserts.
Air volume 1000 m3/h.

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1.2.24 OPTIONAL EQUIPMENT FOR CNC CONTROL

- 1.2.25 1 EMAG REMOTE EXPERTS
Price per year 2.500,00 €
(EMAG Remote Diagnosis included)
CNC control Sinumerik 840Dsl with PCU50 and HMI-Operate
- Assistance key
- 24/7 telephone support via our service hotline
- Quarterly documentation on machine evaluations and breakdowns
(reports on efficiency, reliability, connection monitoring, service requests, etc.)
- If required, consultation with remote experts, expert service worldwide
(Mo - Fr 0700h - 1700h GMT + 0100)
- 1.2.26 1 NETWORK DRIVE CONNECTION
(Control system Siemens 840Dsl)
Linking the machine control to your network after startup in your factory, so that network drives are available in the user interface. NC-programs or any data can be stored or loaded from there.
Customer-specific requirements / restrictions for network integration:
- Operation with DHCP server yes / no; if no, then IP address, sub-net mask and default gateway
- DNS server and enable code of network source (also IP-address of server, if DNS not known)
- User name and password for the automatic preparation of the connection
- Network cable into the electrical cabinet
When providing server and directory names it has to be taken into account that the 840Dsl features an English language Windows operating system that cannot handle the German umlauts ä, ö and ü.
When installing of security update, the company Siemens restrictions are to consider. If in the domain automatically software is installed on all computers, so the functionality is no longer guaranteed after restart!
- 1.2.27 1 EMERGENCY RETRACTION AT POWER FAILURE
Interpolated retraction of axis in case of power failure, voltage fluctuation and drive failure. Electric components (condenser modul) are used which maintain the voltage for the duration of the retreat and use the regenerative energy for the drive autonomously retract of the axes.
(strongly recommended for unstable power supply, expensive tools or workpieces)
- 1.2.28 1 INTERFACE TO EXTERNAL HANDLING SYSTEM (LH), SIEMENS
(turnover station, transfer unit, robot with independent control)
Safety signal interface via HAN24E plug on electrical cabinet
with 4 potential-free inputs and 4 potential-free outputs
for the change of safety signals "emergency-stop" and "access door open"
Handling-signal interface via HAN16E plug on electrical cabinet
(Located on the right side of the machine)
Deviations of the standard interface on request at additional costs.
(1.000,00 € / signal)

Price does not include options (*)

Offer : 160246 / 4

Hyundai WIA Corp.

31/03/2017

11/28

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- 1.2.29 1 INTERFACE TO EXTERNAL HANDLING SYSTEM (RH), SIEMENS
(turnover station, transfer unit, robot with independent control)
Safety signal interface via HAN24E plug on electrical cabinet
with 4 potential-free inputs and 4 potential-free outputs
for the change of safety signals "emergency-stop" and "access door open"
Handling-signal interface via HAN16E plug on electrical cabinet
(Located on the right side of the machine)
Deviations of the standard interface on request at additional costs.
(1.000,00 € / signal)
- 1.2.30 1 WORKSHOP-ORIENTATED PROGRAMMING AC AND UF-JOINTS
For outer races (straight grooves).
For quick and easy programming parameters for componentrelated dimensions are
adapted.
Same process for tool-offset.
Every tool is allocated an empirically determined preset quantity of workpieces to be
machined.
On expiry of the tool life the machine is stopped.
Control monitor display: tool-no., preset and remaining tool life quantity of workpieces per
tool.

Price does not include options (*)

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12/28

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1.2.31 OTHER AUXILIARY EQUIPMENT

- 1.2.32 1 SPARE PARTS PACKAGE
Budgetary amount for spare parts package
(Preliminary, final definition and list of selected components will be forwarded in case of an order)
This package is fixed priced, non discountable.
- 1.2.33 1 ASIAN PACKAGE
for ambient temperature of max. 55°C at max. 95% humidity
voltage max. 415V $\pm$ 10%
additional heating for operator panel, cooling from hydraulik oil under 60°C
additional cooling for operating panel.
- 1.2.34 1 3-COLOUR STATUS LIGHT (Q-LIGHT)
mounted on machine. Colour code:
red: machine stopped
yellow: warning -tool to be changed,
-lubricant running low, etc.
green: machine in automatic cycle.
- 1.2.35 1 SPECIAL MACHINE COLOUR
Complete machine according to customers colour specification.
Like noticed in "Specification for Quotation" p.t. 16. Color.

Price does not include options (*)

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1.3 COMPONENT RELATED EQUIPMENT

1.3.1 TOOL EQUIPMENT

- 1.3.2 1 SET OF MILLING TOOLHOLDERS HSK 63-C FOR ROUGH MILLING
- 3 pcs toolholders for interchangeable milling heads
- 1.3.3 1 SET OF MILLING TOOLHOLDERS HSK 63-C FOR FINISH MILLING
- 12 pcs toolholders for interchangeable milling heads.
- 1.3.4 1 SET OF MILLING TOOLS FOR SOFT MILLING OF OUTER RACE AC21021-C8001
Head system with carbid cutting inserts
consisting of:
- 30 cutting heads for rough milling
- 30 cutting heads for finish milling
Please consider that EMAG can not give a guarantee for the tool life.
- 1.3.5 1 SET OF MILLING TOOLS FOR SOFT MILLING OF OUTER RACE AC21022-C8001
Head system with carbid cutting inserts
consisting of:
- 30 cutting heads for rough milling
- 30 cutting heads for finish milling
Please consider that EMAG can not give a guarantee for the tool life.
- 1.3.6 1 SET OF MILLING TOOLS FOR SOFT MILLING OF OUTER RACE AC21023-C8001
Head system with carbid cutting inserts
consisting of:
- 30 cutting heads for rough milling
- 30 cutting heads for finish milling
Please consider that EMAG can not give a guarantee for the tool life.
- 1.3.7 1 SET OF MILLING TOOLS FOR SOFT MILLING OF OUTER RACE AC21024-C9002
Head system with carbid cutting inserts
consisting of:
- 30 cutting heads for rough milling
- 30 cutting heads for finish milling
Please consider that EMAG can not give a guarantee for the tool life.
- 1.3.8 1 EQUIPMENT FOR BALL TRACK MILLING MACHINE
consisting of:
- 1 cone wiper
- 1 torque wrench with insert
- 2 plugs HSK63
- 1 ball plug gauge HSK63
- 1.3.9 1 SETTING DIE FOR MILLING HEAD EXCHANGE
consisting of:
- 1 table-mount fixture for tool change

Price does not include options (*)

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31/03/2017

14/28

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1.3.10 CHUCKING SYSTEM

- 1.3.11 2 ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER SIN85
- with 1 medium duct
Max. drawbar force 3.325 daN at 70 bar.
The clamping pressure is manually infinitely adjustable. .
The clamping cylinder stroke and the clamping pressure are controlled by a linear measuring system and by a pressure switch.
- 1.3.12 2 WORKPIECE POSITION MONITORING, (AIR SENSING)
The position of the workpiece in the clamping fixture is monitored by measuring the air pressure between the part stop and the component. This establishes contact accuracy. Inaccuracies on the workpiece resulting from deformation of clamping fixture or workpiece etc., are not recognised. The air is supplied by one or more nozzles (depending on clamping fixture type). The pressure is measured with a pressure switch.
The air supply is controlled by a directional valve.
The air flow is cleaned by a service unit and sent through the main spindle.
NOTE:
Requires CHUCKING CYLINDER WITH AIR THROUGH THE SPINDLE.
Equipping the clamping fixture with a pneumatic system monitor is not included in the price of this item.
- 1.3.13 2 COLLET CHUCK 120 (HAINBUCH)
Outer dia. 235 mm
With basic stop.
Clamping range 60 -120 mm.
Prepared for air sensing.
Without workpiece-specifically interchangeable parts.
- 1.3.14 2 SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (HAINBUCH) AC 21021-C8001
Laid out for one Outer Race.
Consisting of clamping head, part stop and control ring.
- 1.3.15 2 SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (HAINBUCH) AC 21022-C8001
Laid out for one Outer Race.
Consisting of clamping head, part stop and control ring.
- 1.3.16 2 SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (HAINBUCH) AC 21023-C8001
Laid out for one Outer Race.
Consisting of clamping head, part stop and control ring.
- 1.3.17 2 SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (HAINBUCH) AC 21024-C9002
Laid out for one Outer Race.
Consisting of clamping head, part stop and control ring.

Price does not include options (*)

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15/28

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- 1.3.18 1 CHANGING FIXTURE FOR COLLET CHUCK 120
for quick changing of the clamping head.
Pneumatically operated.

Price does not include options (*)

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31/03/2017

16/28

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1.4 SERVICES

- 1.4.1 1 Please note that no additional services are contained in the previously specified prices. We would be pleased to carry out the following services and would quote according to your wishes.
- Preparation of tooling layouts and NC programs
 - Set-up machine for component trials on your specific workpiece in our works
 - Machine approval in our works by your engineers.
 - Final acceptance at customer's plant with a specific workpiece
- 1.4.2 1 PROJECT ENGINEERING
Project engineering and coordination with suppliers and customers.
Engineering investigations (machine set up, with central automation, central exhaust, chip disposal)
Customization of interfaces with the suppliers.
- 1.4.3 1 TOOLING LAYOUT FOR OUTER RACE AC21021-C8001
Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.
- 1.4.4 1 TOOLING LAYOUT FOR OUTER RACE AC21022-C8001
Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.
- 1.4.5 1 TOOLING LAYOUT FOR OUTER RACE AC21023-C8001
Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.
- 1.4.6 1 TOOLING LAYOUT FOR OUTER RACE AC21024-C9002
Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.
- 1.4.7 1 MACHINE SETUP FOR OUTER RACE AC21021-C8001
Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance.
Tool life optimisation not included.
- 1.4.8 1 MACHINE SETUP FOR OUTER RACE AC21022-C8001
Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance.
Tool life optimisation not included.
- 1.4.9 1 MACHINE SETUP FOR OUTER RACE AC21023-C8001
Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance.
Tool life optimisation not included.

Price does not include options (*)

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- 1.4.10 1 MACHINE SETUP FOR OUTER RACE AC21024-C9002
Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance.
Tool life optimisation not included.
- 1.4.11 1 PRE-ACCEPTANCE OF THE MACHINE
Specific workpiece: **AC 21024-C9002.**
including measuring of workpiece at EMAG (3D measuring machine, FAG roundness measuring device, etc.).
One workpiece in one clamping operation. Max. 30 components.
Submission of proof to Cm of machine's capability of max. 3 measuring characteristics.
For Cm of machine's capability measurements of workpiece 1, 5, 10, 15, 20, 25, 30.
- 1.4.12 1 INSTALLATION AND DEMONSTRATION (OVERSEAS)
in your works
- The following services are included:
- Machine installation and demonstration
 - Machine approval and hand-over
 - Talk-through of the machine documentation and service instruction
 - Operator instruction
 - Travel costs
- Total of 15 man days
- Not included are:
- Machine set-up
 - Machine operation for production
 - Additional service
- Additional service engineer costs for time in excess of that quoted will be charged extra.
- 1.4.13 1 FINAL ACCEPTANCE AT CUSTOMER'S PLANT AC21021-C8001
Specific workpiece: **AC21021-C8001.**
Gauging of the workpieces with measuring equipment supplied by customer.
Uninterrupted machining of 30 workpieces
Evaluation and guaranteed criteria are the same as for pre-acceptance test.
- 1.4.14 1 FINAL ACCEPTANCE AT CUSTOMER'S PLANT AC21022-C8001
Specific workpiece: **AC21022-C8001.**
Gauging of the workpieces with measuring equipment supplied by customer.
Uninterrupted machining of 30 workpieces
Evaluation and guaranteed criteria are the same as for pre-acceptance test.

Price does not include options (*)

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- 1.4.15 1 FINAL ACCEPTANCE AT CUSTOMER'S PLANT AC21023-C8001
Specific workpiece: **AC21023-C8001.**
Gauging of the workpieces with measuring equipment supplied by customer.
Uninterrupted machining of 30 workpieces
Evaluation and guaranteed criteria are the same as for pre-acceptance test.
- 1.4.16 1 FINAL ACCEPTANCE AT CUSTOMER'S PLANT AC21024-C9002
Specific workpiece: **AC21024-C9002.**
Gauging of the workpieces with measuring equipment supplied by customer.
Uninterrupted machining of 30 workpieces
Evaluation and guaranteed criteria are the same as for pre-acceptance test.
- 1.4.17 1 INSTALLATION POINT
The installation point has to be prepared in accordance with the documentation supplied.
We take it for granted that all conditions are met before our installation crew arrives.
Where waiting times are incurred due to the conditions not having been met, these will be charged extra. During installation and commissioning our engineers have to be provided, on request, with the support of skilled customer personnel and similar assistance. This also applies to the operation of cranes, handling equipment and other aids.
- 1.4.18 1 MACHINE LIFTING EQUIPMENT
EMAG will provide a lifting equipment to lift the machine by crane, free of charge.
However, should the customer not return the equipment to EMAG within 4 weeks, freight and charges prepaid, a separate invoice of EUR 1.500,00 will be issued.
- 1.4.19 1 PRODUCTION SUPPORT 1 WEEK AFTER START OF PRODUCTION
1 week production support at customers plant
Performed by EMAG India.

Price does not include options (*)

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19/28

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1.4.20 DOCUMENTATION

1.4.21 1 DOCUMENTATION (Siemens)

OPERATING INSTRUCTIONS

1 complete set of operating instructions on paper

Language: English

Manuals for the machine:

- Introduction
- Safety instructions
- Product description
- Transport and installation
- Start-up
- Production
- Preventive maintenance

Manuals from the manufacturer of the control unit:

Siemens SIN 840D sl

- User manual
- Diagnostics manual
- Programming - Basics
- Programming - Cycles
- Programming - Operations planning and scheduling

Machine diagrams:

- Machine layout
- Transport diagram
- Machine data sheet
- Drive data sheet
- Connection data sheet for fluidics
- Connection data sheet for electrics

List of spare parts and wearing parts

Operating instructions from third parties

Documentation for clamping devices and tools

TECHNICAL DATA

1 set of technical data on paper, German or English

- Electrical circuit diagram with list of el. devices
- Fluid system diagram with list of el. devices
- Drawings of subassemblies with relevance for spare parts
- List of components with relevance for spare parts and wearing parts in relation to the drawings
- Technical data, test reports, declarations of incorporation from third parties

Other languages available on request for an additional charge; third party documentation excepted

1.4.22 5 ADDITIONAL SET OF DOCUMENTATION

1 additional set of the documentation described above.

Language: English 4x, Korean 1x (same scope as delivered)

Price does not include options (*)

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- 1.4.23 2 **DOCUMENTATION ON CD**
1 complete set of documentation on CD

Price does not include options (*)

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31/03/2017

21/28

**1.4.24 CUSTOMER SPECIFICATION**

- 1.4.25 1 COMPLIANCE SPECIFICATION SHEET / DELIVERY CONDITIONS
Design of mechanical-, electrical- and fluid equipment according to "M/C Specification,
Date: 24th January 2017, Rev. 1"
Due to our design we are unable to comply with the following items of your specification
sheet, meaning we must apply for an exceptional permission.

Item	Customer requirement	EMAG commitment
Basic Condition	AC 415V (+/-15%) x3 phase x 50Hz	AC 415V (+/-10%) x3 phase x 50Hz like in your specifications item Electric 1. Main Circuit.
Noise level	less than 75 dB (A)	up to 80dB (A)
General specification of Hyundai WIA	Keep the "General specification of Hyundai WIA"	This document is not part of the inquiry and therefore it is not considered
Durable & Perishable Tool	For 180.000 pcs. of workpiece machining (Turning and milling insert)	Quantity has changed compared to the RFQ (see separate position)
Main Circuit	Solenoid valve: 115V AC 50Hz, Lamp Type, Communication Type	24V DC with LED
CNC & PLC	- Voltage 240V - Memory USB Card - Memory EPROM	400V AC / 24V DC - Memory USB-Stick - No longer available from Siemens.
CNC & PLC	Network interface - LAN	USB
Wiring	Remain the 30% marginal space and do not wire it hose together	max. 10% possible
Items for the capability (pre- and final acceptance)	Ball Track Milling: Groove Dia, Center, Ball Diameter	Ball diameter is no item for evaluation (Cm)
Detail 15. Others	33) Manuel mode on the machine, in / out conveyor should be possible for automated mode. 34) The movement of stocker should be possible for forward / backward	- Our automation is controlled by the machine. No separate control for automation. Because of that it's not possible. - The storage units are operated with hinge straps. Because of that is a backward movement not possible.

Price does not include options (*)

Offer : 160246 / 4

Hyundai WIA Corp.

31/03/2017

22/28

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1.4.26 TRAINING

- 1.4.27 1 OPERATING COURSE (Basic training)
For two participants in your works, performed by EMAG India .
Matched to the machine design (modular customized)
Training course duration: 3 days / shift (shift monday - friday).
Language: English
Documentation: 4x English

- 1.4.28 1 MECHANICAL SERVICE COURSE (Basic training)
For two participants in your works, performed by EMAG India.
Matched to the machine design (modular customized)
Training course duration: 3 days / shift (shift monday - friday).
Language: English
Documentation: 4x English

1.4.29 TRANSPORT COSTS

- 1.4.30 1 PACKING AND TRANSPORTATION OF TEST PIECE AND INSPECTION GAUGE
FROM INDIA
Transport from customers plant in India.
- 1.4.31 1 PACKING AND TRANSPORTATION OF TEST PIECE AND INSPECTION GAUGE
FROM KOREA
Transport from customers plant in Korea.
- 1.4.32 1 FOB DELIVERY FREE GERMAN SEAPORT
including seaworthy packing.

Price does not include options (*)

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1.5 MACHINE SPECIFICATIONS

1.5.1 1 TECHNICAL MACHINE DATA VSC 315 DUO KBU Subject to change without notice

WORKING AREA

Chuck diameter	mm	250
Swing diameter	mm	260
X-stroke	mm	900
Y-stroke	mm	315
Z-stroke	mm	300

WORKPIECE DIMENSIONS

Workpiece length, max.	mm	260*
Workpiece O/D, max.	mm	120
Ball section dia., max.	mm	30

(*Longer workpieces on request)

MAIN SPINDLE

Spindle flange acc. to DIN 55026	size	8
Spindle front bearing diameter	mm	140
max. spindle speed	rpm	3000

MAIN DRIVE

AC-Synchronous motor 40/100 % ED	kW	28,5/27,5
Max. power	kW	28,5
Full power at spindle speed	rpm	700
Torque at 40/100 % ED	Nm	590/375
Max. torque at	Nm	590

FEED DRIVE

Rapid traverse feed, X+Y	m/min	45
Rapid traverse feed, Z	m/min	30
Feed force, X/Y/Z	kN	11
Ball screw diameter X/Y/Z	mm	50/40/40

MILLING SPINDLE

AC-motor	kW	15
Max. spindle speed	rpm	6.000

ELECTRICAL EQUIPMENT according to VDE 0113

Supply voltage	V	400
Control circuit voltage - DC	V	24
- AC	V	230
Frequency	Hz	50
Total connected load	kVA	120
Input fuse	A	250

DIMENSIONS AND WEIGHT

Length	mm	4400
Width	mm	3600
Height	mm	3500
Weight	kg approx.	20000

1.5.2 1 SIEMENS OPERATOR PANEL 12" TFT COLOR MONITOR consisting of OP012 + PCU50 + Control panel (MCP483)

Price does not include options (*)

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24/28

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- 1.5.3 1 CNC CONTROL SIEMENS
SINUMERIK 840 Dsl with integrated PLC S7-300
Operator software Sinumerik Operate with HMI-Pro (Windows 7)
Workstation electrically interlocked with UPS (Uninterruptible Power Supply)
Screen language and on-screen messages in German / English / in official EU language,
more languages available as an optional
CNC program and data memory 3.0 Mbyte, up to 9 MB additional memory is available as
an option.
USB-interface (front) for memory-stick.
Machine diagnostics via internet with VPN technology to verify EMAG's standard format
Software backup is to external media (USB stick integrated into PCU50) to verify EMAG's
standard format
dataGrabber (Comara) – Capable of recording of process data (raw data) for diagnostics
and process optimization – System is not functional without evaluation software.
Optional tool monitoring:
- Toolinspect (turning, milling), - Zinngrebe (only VLC + grinding)
EMAG ADD-ON FUNCTIONS:
Machine-specific documentation of functions and parameters is available directly on the
control unit. (PDF display)
Automatic return to the home position upon machine error. Error recovery is a simple one
button process.
First run high speed traverse is at a reduced speed from home to the first cut. Reduced
speed allows you to avoid a collision and save damaged equipment and reprogramming
costs.
Internal measuring probe with collision monitoring on the traversing movements (Option)
Automatic workpiece gaging when switching to a replacement tool or tool insert (Option)
Automatic filling and emptying of the entire machine.
"Optional Stop" (M01) allows you to make a programmed stop or interrupt production after
finishing a part at a desired point.
Single cycle: Allows one part to be transferred and machined for setup and tryout
purposes.
STRUCTURED WORKPIECE PROGRAMMING:
Division of individual programs into subprograms, e.g. the program for the machining
contains only the technology program and not the loading part.
The loading program has individual parameters. Division into machine and work piece
specific parameters.
Programmable chuck with length measuring system in the pull rod.
Different clamping situations (Unclamped, Clamped and Over stroke) can be changed by
parameter assignment. Displayed by diagnostics menu.
Recording and displaying of the most important production data: work piece counter, shift
counter, cycle time, work piece status.
Production of definable batch sizes. The machine stops upon reaching the selected no.
Optimized programming for the turret keeps chips from collecting.
Rinsing of clamping devices after a programmable number of parts
Number and positions of the workpiece carriers can be defined for closed automation.
Lifecycle screen that displays tool wear and tool life.

Total:	1.223.850,00 €
Discount:	373.850,00 €
Final Price:	850.000,00 €

Price does not include options (*)

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31/03/2017

25/28



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Commercial Conditions

CONFIDENTIALITY

Any commercial and operational written or oral information or data relating to this quotation / order, such as drawings, specifications, recommended machining data and clamping methods, etc., shall not be divulged to any third party (referring, in particular, but not exclusively, to competitors, for the purpose of bid comparisons) and are to be used solely within the meaning of this quotation / order.

The recipient of this quotation / order expressly accepts these obligations.

MACHINING

The determined cycle times result from values, which are based on data and experiences of acknowledged cutting material manufacturers. Changing rough part dimensions, tolerances, material as well as the material hardness are only possible during building and testing of the machine. Changes will only be accepted in written form and in mutual agreement.

PROVISION OF EQUIPMENT AND MATERIALS BY CUSTOMER

All tooling, work holding equipment, parts, etc. are to be supplied with DDP supplying plant EMAG Incoterms 2010. We wish to point out that we accept no responsibility for the functional capability and working order of any equipment provided.

The functional capability of the provided equipment is the responsibility of the buyer.

In the event of problems, we assure you of our constructive collaboration. Any additional costs arising from modification, optimization, etc. will be charged to the customer. Furthermore, we do not guarantee any quoted machining times in such cases.

When orders cannot be carried out on schedule, caused by material supplied by the customer or if technical problems should occur, the machine will be invoiced on the originally agreed delivery date, when the relevant payment will become due.

The agreed upon date of delivery date may be postponed to the time to achieve full operability, which shall not result in any event of default of the manufacturer.

DELIVERY PERIOD

FOB 10.02.2018 german seaport. The delivery period is not binding and can change between the submission of quotation and placement of order.

Please discuss the subject of delivery with us at time of order placement and we will do everything possible to meet your requirements. Changes regarding the component/s or the machines or the order content after order placement may result in changes in price and delivery.

Compliance of delivery date requires timely down and / or partial payment by the customer within the prescribed payment terms. If receipt of down and or partial payment is delayed, the delivery shall be postponed accordingly, which shall not result in any event of default of the manufacturer.

CONFORMITY WITH CUSTOMERS` REGULATIONS

has not been taken into account. Where unavoidable deviations from such regulations prove necessary, specific authorization will be requested.

PRICES

FCA (Free Carrier, Incoterms 2010). Packing, insurance and transport not included.

The payments are to be made without deduction. With delayed payment the relevant laws apply.

VALIDITY OF PRICES

The prices quoted are valid for 90 days from the date of quotation. Orders placed after this period would be subject to a price increase of 0.5 % per month to cover increases in labour and material costs.

This quotation will replace all previous versions.

Price does not include options (*)

Offer : 160246 / 4

Hyundai WIA Corp.

31/03/2017

26/28

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TERMS OF PAYMENT

20% of total amount: at the time of P/O (by T/T)
70% of total amount: FOB (by irrevocable L/C)
10% of total amount: Final acceptance (L/C or T/T)

All L/C-charges are for buyer's account. L/C must be opened latest 30 days after order confirmation

RETENTION OF TITLE

The machine will remain the property of the builder until complete payment is received from the buyer. Upon placement of order the extended retention of title according to clause 48 of our General Conditions for Sales and Deliveries shall be deemed as agreed.

LIMITATION OF LIABILITY

Any claims for compensation of the buyer including but not limited to claims resulting on damages not incurred on the delivery item itself (indirect or consequential damages such as loss of profit, loss of production) are limited to 20% of the value of the delivery item. This liability cap shall not apply in cases of a wilful misconduct of the manufacturer or any other liability by mandatory law.

PERIOD OF LIABILITY

Any rights and claims of the buyer caused on defects shall be subject to a limitation period of twelve months.

The limitation period starts with the final acceptance, but no later than 3 months from delivery, and does not apply to wear and tear parts.

WORK PIECES FOR MACHINE ACCEPTANCE

Both, pre-acceptance and final acceptance tests are carried out with either an EMAG test component to VDI standard, or the work piece specified by the customer's order.

The customer will supply an agreed number of work pieces for machining process optimisation and pre-acceptance tests. These work pieces must be from a single batch and of a quality that corresponds with the relevant manufacturing drawing. For preparatory work and preliminary machine acceptance the customer will, at the agreed time, supply EMAG free of charge with a sufficient number of sample work pieces of a quality that corresponds with the relevant manufacturing drawing. Any delay in the supply of these work pieces will result in the machine delivery being delayed and the starting date of a possible delayed machine delivery penalty being deferred accordingly.

FINAL ACCEPTANCE AND FINAL PAYMENT

Final acceptance tests are carried out as soon as the machine has been installed and commissioned.

For this purpose, the buyer has to provide a sufficient number of work pieces, personnel and capacity to measure and evaluate the acceptance components.

Once the machine is taken into production and the agreed component quality and cycle time have been achieved, final acceptance is considered to have taken place.

Where final acceptance of the machine is not possible owing to reasons beyond EMAG's control (missing acceptance components, etc.), final contractual payment falls due immediately.

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27/28

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TECHNICAL AVAILABILITY

Our confirmation of the technical availability according to VDI 3423 is based on following conditions:
Exclusively qualified operating personnel, monthly availability protocols addressed to our service manager, demonstrable preventive maintenance is carried out according to our machine documentation, connection to our tele service system if requested by our service department.
In case of non-compliance of these conditions, our commitment expires.

STANDARDS AND REGULATIONS

EMAG Machines are built in accordance with the new EC machine guidelines 2006/42/EG and follow the standards and safety regulations. The machine carries the EC-sign that agrees with the guidelines and receives the EC conformity.

TECHNICAL DEVELOPMENT

In the interest of technical progress, we reserve ourselves the right to make design changes which are due to continuous product development.

RESERVATION CLAUSE

The equipment quoted herein may be subject to German and/or EU export regulations. Therefore, an end user statement may be required for approval by German authorities (BAFA), stating the intended use of the equipment.
This offer is subject to the control and to the granting of an export license by the German Federal Office of Economics and Export Control (BAFA).

ADDITIONAL PROVISIONS

In addition and subordinated to this quotation our General Conditions for Sales and Deliveries for International Business shall apply (download under http://www.emag.com/Delivery_conditions_UN.pdf).

This document was created electronically and is valid without signature.

Yours faithfully,

EMAG Maschinenfabrik GmbH

Bernhard Böttigheimer

Do-Youn Koo
Area Sales Manager

Enclosures

EMAG-General conditions for sales and deliveries (UN)
Layout
Notes on specifications
Timestudy

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